

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (EC)

Nov 2012

EC312 Digital Signal Processing

Date: 09.11.2012, Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) List out the applications of DSP in various fields. Differentiate the continuous valued and discrete valued signals. [05]

(b) Determine causality and time variance properties of $y(n) = \sum_{k=-\infty}^n x(k)$ [02]

Q - 2 (a) Describe the properties of linear convolution by taking appropriate examples. [05]

(b) List out the properties of Z-transform and prove any two with example. [05]

(c) Determine the inverse Z-transform of $X(z) = \frac{z^2}{(0.5 - 1.5z + z^2)}$ [04]

For ROC $|z| < 0.5$ Using long Division Method.

OR

Q - 2 (a) Explain the Gibb's phenomenon for FIR filter designing. [05]

(b) What is circular convolution of DFT? Explain with example. [05]

(c) List out the properties of DFT and prove any one. [04]

Q - 3 (a) Explain sampling theorem and For an analog signal, $x(t) = 3\cos 50\pi t + 10 \sin 300\pi t - \cos 100\pi t$, calculate the Nyquist sampling rate. [04]

(b) Find the convolution using mathematical method for system which having impulse response $x[n] = \{1, 2, 1, 2\}$ and input $h[n] = \{1, 1, 1\}$ [06]

(c) Explain minimum - phase systems. [04]

OR

Q - 3 (a) Compare the characteristics of IIR and FIR filter and Draw the parallel realization of filter [06]

$$H(z) = \frac{1 - z^{-1}}{1 - 0.2z^{-1} - 0.15z^{-2}}$$

- (b) Compute the 8- point circular convolution for following sequences [05]

$$x_1(n) = \{1, 1, 1, 1, 0, 0, 0, 0\}$$

$$x_2(n) = \sin\left(\frac{3\pi n}{8}\right) \quad 0 \leq n \leq 7$$

- (c) Explain Signal ambiguity in frequency domain (aliasing). How this aliasing effect is removed? [03]

SECTION - II

- Q - 4 (a) Give the importance of "Zero Padding" in the discrete Fourier transform computation. [02]

- (b) Determine the length-4 sequence from its DFT [05]

$$X(k) = \{4, 1-j, -2, 1+j\}$$

- Q - 5 (a) Obtain direct form I and direct form II realization of a system described by, [07]

$$y(n) - \frac{3}{4}y(n-1) + \frac{1}{8}y(n-2) = x(n) + \frac{1}{2}x(n-1)$$

- (b) Draw and explain radix 2 decimation in time FFT implementation signal flow graph (butterfly diagram) for 8 point FFT. [07]

OR

- Q - 5 (a) Consider a causal LTI system with system function, [07]

$$H(Z) = \frac{1 + \frac{1}{5}Z^{-1}}{\left(1 - 0.5Z^{-1} + \frac{1}{3}Z^{-2}\right)(1 + 0.25Z^{-1})}$$

Draw the signal flow graph for implementation of the system using direct form-II

- (b) (i) Two signals $x_1(t) = \cos 20\pi t$ and $x_2(t) = \cos 100\pi t$ are sampled with sampling frequency 40Hz. Obtain the associated discrete time signals $x_1(n)$ and $x_2(n)$. [07]

(ii) Compute the cross correlation between

$$x[n] = \{1, 1, 0, 1\} \quad y[n] = \{4, -3, -2, 1\}$$

- Q - 6 (a) Determine the homogeneous solution of the system described by [07]

$$y(n) - 3y(n-1) - 4y(n-2) = x(n)$$

- (b) Write Short note on Goertzel Algorithm. [07]

OR

Q - 6 (a) Determine the Z- transform and sketch the ROC of

[07]

$$1) \quad x(n) = \begin{cases} \left(\frac{1}{3}\right)^n & n > 0 \\ \left(\frac{1}{2}\right)^{-n} & n < 0 \end{cases}$$

$$2) \quad x(n) = \left(\frac{1}{2}\right)^n \{u(n) - u(n-10)\}$$

(b) Given $x(n]=2^n$ and $N=8$, Find $X(k)$ using DIT FFT algorithm.

[07]
